Real course photos



Figure 1: view from start



Figure 2: View from first turn

Real course details

First hallway section width: 84”
Second hallway section width: 92”
Left-side inset -> second hallway section start: 129”
Left inset depth: 26.8”
Left inset length: 49.8”

Bin length (top cross-section): 20”
Bin width (top cross-section): 10.8”

Left-side inset -> bottom edge of first pair of bins: 32”
Right wall -> top right edge of first pair of bins: 58.5”
Front edge of first pair of bins -> back edge of second pair of bins: 37”
Left wall -> top left edge of second pair of bins: 57.8”
Front right wall corner -> front right edge of third bin: 57”
Front left edge of third bin -> left wall: 35”

Proven to be comfortably manually driveable via teleoperation. Requires tight turning.

Run #1

bag name: real_bag1
The robot made it past the first obstacle but stopped after it cleared it. The robot was then switched to manual teleoperation mode and driven back to the start. We noticed that the costmap had incorrectly marked some cells as being occupied when they weren’t - you can see that as the tester drove the robot back; I did not hit anything in real life but passed right through occupied costmap cells.

Run #2

bag name: real_bag2
The robot successfully cleared the course; this is the intended behavior. If only it could do this every time…

Run #3

bag name: real_bag3
Robot stopped in the same spot as bag #1. This run had a different starting location.
After it stopped, the tester walked up to it and stopped the bag.

Run #4

bag name: real_bag4

Once the waypoint was given, the tester had to walk around the robot to create “ghost obstacles” to prevent it from planning incorrectly. This is normal; in real operation, the robot’s lane detection will prevent this from happening. After it began navigation, it stopped in the same spot as runs #1 and #3, was switched to teleoperation mode, manually driven back to the start, and the bag was stopped.